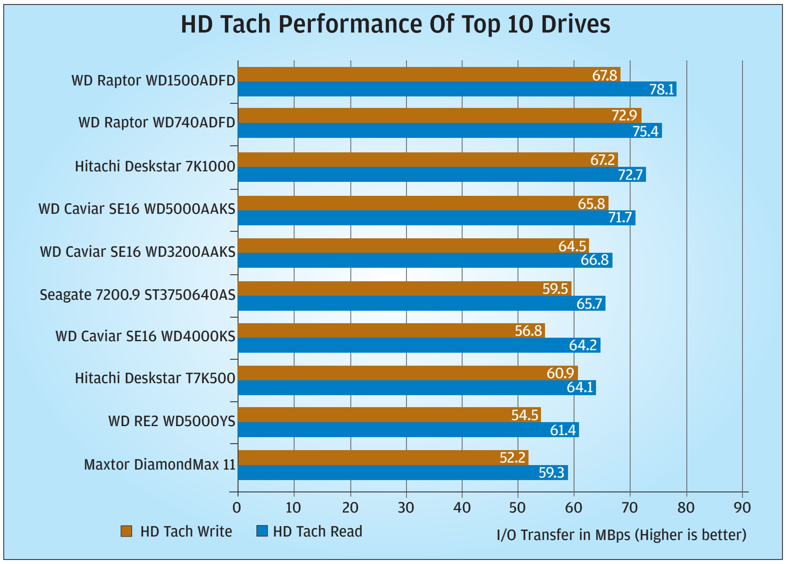

found to have increased tremendously, and the credit goes to the superior SATA controller on the Intel 965G chipset-based motherboard which we used this time round. The Raptors boasted the fastest speeds, thanks to their 10K rpm. The Raptor WD1500ADFD topped the charts with average read of 78.1 MB/s in the average write test; the other Raptor, the WD740ADFD, top-scored with 72.9. Next in line were the Hitachi Deskstar 7K1000 and the WD Caviar SE16 WD5000AAKS with very good performance.

CPU utilisation and burst speed do not have much importance these days: CPU utilisation does not matter much because we have very powerful processors even in entry-level PCs these days, and burst speed has little significance as it only indicates the absolute maximum level of data throughput, which is almost never encountered in real-life scenarios. CPU utilisation of most of the drives hovered between 2 and 4 per cent, which is not much. The burst rate of the Seagate 750 GB drive was the highest at 248.7 MBps, again a step higher than what it did last year, probably due to the better SATA controller.

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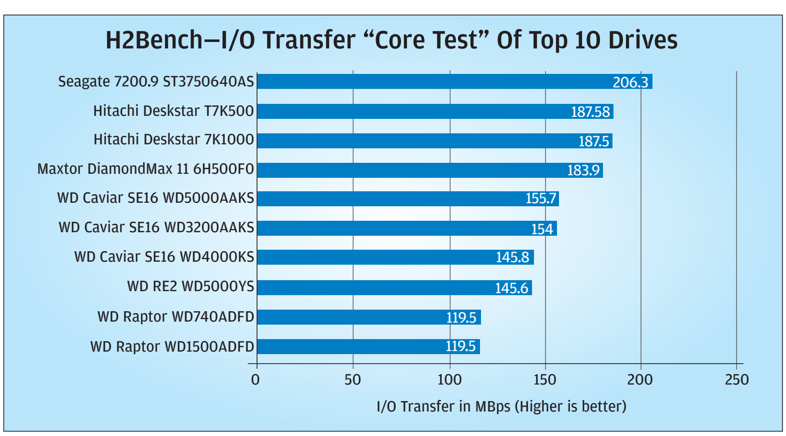
Speed can come cheap!

H2Bench

This is a new benchmark. It measures drive performance at a low level and returns a very precise measure of the performance. Like with HD Tach, in H2 Bench too, the Seagate 750 GB trounced the rest and posted the highest scores in the repetitive sequential read or "Core Test," which we conducted at 50 per cent of the drive capacity. It posted an impressive 206.3 MB/s, followed by the two Hitachis with almost identical scores of around 187.5 MBps. But when it came to read and write random access times, the WD Raptors once again came forth with the highest scores—quicksilver response times. Lower read and write access times means the drive should be able to speedily handle assorted data.



PCMark05

We saw mixed results in the PCMark05 test. While the WD Raptors scored very well in the XP Startup as well as the Application Loading tests, the SATA II drives finally started proving their prowess by emerging with the highest scores in the Virus Scan part of the test. The Maxtor DiamondMax 11 was the top scorer here, while the two Hitachis followed it closely. For some unknown reason, the Samsung HD160JJ 160 GB failed to complete this test, while the Seagate Barracuda 7200.9 ST3500641AS could not complete the Virus Scan part. We tried to repeat the test several times, but to no avail.

SiSoft Sandra Pro Business 2007

We saw four top contenders after conducting the SiSoft Sandra benchmarks. The usual suspects, the WD Raptors, were at the top, but very close to them this time were the Hitachi Deskstar 7K1000 and the WD Caviar SE16 WD5000AAKS, all with respectable scores breaching the 70 MBps barrier in the drive index. The scene was similar when we took a look at the sequential and assorted read and write indices.

The drive index is derived from the combination of the random and sequential read and write indices. It is therefore a measure of the drive's transfer rate, independent of the interface.

SiSoft Sandra also measures the access time of the drives, and the results yielded were not surprising. Both the Raptors were yet again the top scorers, along with both the Hitachis and the WD RE2 WD5000YS, with each drive exhibiting an access time of 6 ms.

2. The Real-world Tests

To find out whether the drives match up to the synthetic performance when they are put to work in the real world, we conducted a battery of real-world tests.

File Transfer

We transferred 4 GB of data, sequential and assorted, to and from the drive and also with-